



Software requirements Specification

For

Employee Management System

Software version

Prepared by

Student Name: D.A. Wickramasooriya

Nic:200651802829

Instructor: Mr. Anjana Samarakoon

Course: Web programming II

Batch: Athina Uni

Teaching Assistant: Mr. Anjana Samarakoon

1. Introduction

1.1 Overview: -

- ❖ The Employee Management System (EMS) is a comprehensive web-based application developed to streamline workforce management operations. This system provides functionality for managing employee records, tracking attendance, and handling payroll-related operations. As part of this assignment, an invoicing module has been integrated to demonstrate the use of JavaScript, AJAX, and Hibernate technologies.

1.2 Object: -

- Develop a fully functional web-based Employee Management System with invoicing capabilities
- Implement CRUD operations using Hibernate ORM framework
- Create dynamic user interfaces with JavaScript
- Utilize AJAX for asynchronous server communication
- Ensure data integrity through proper validation mechanisms

1.3 Scope: -

The system encompasses:

- User authentication and authorization
- Employee registration and management
- Invoice generation and management
- Search and filtering capabilities
- Real-time data updates using AJAX

1.4 Technologies Used

Front-end Technologies:

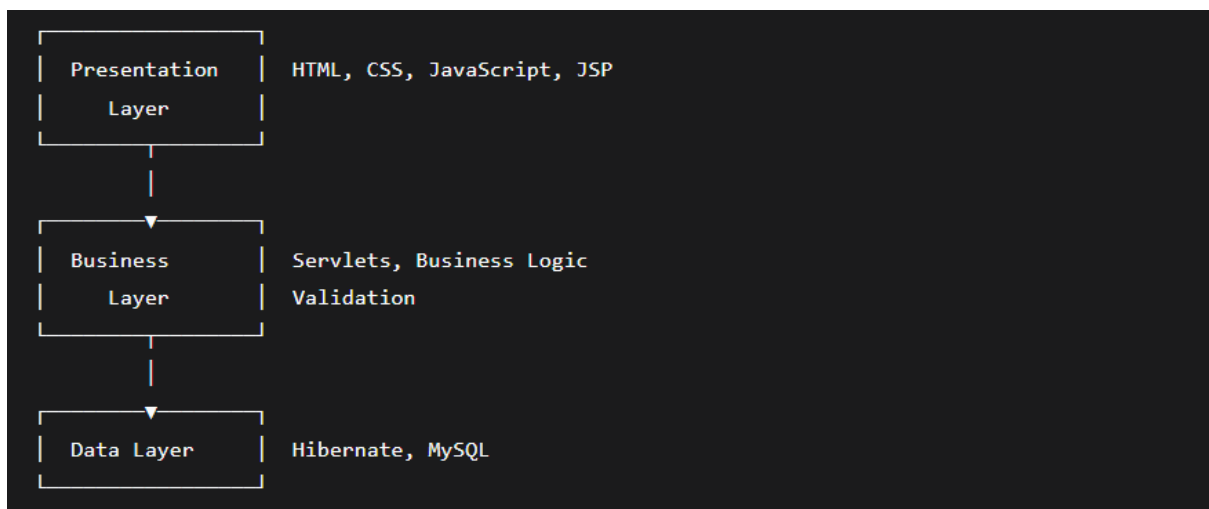
- HTML5
- CSS3
- JavaScript (ES6+)
- AJAX (XMLHttpRequest/Fetch API)
- Font Awesome Icons

Back-end Technologies:

- Java (Servlets, JSP)
- Hibernate ORM Framework
- MySQL Database
- Apache Tomcat Server

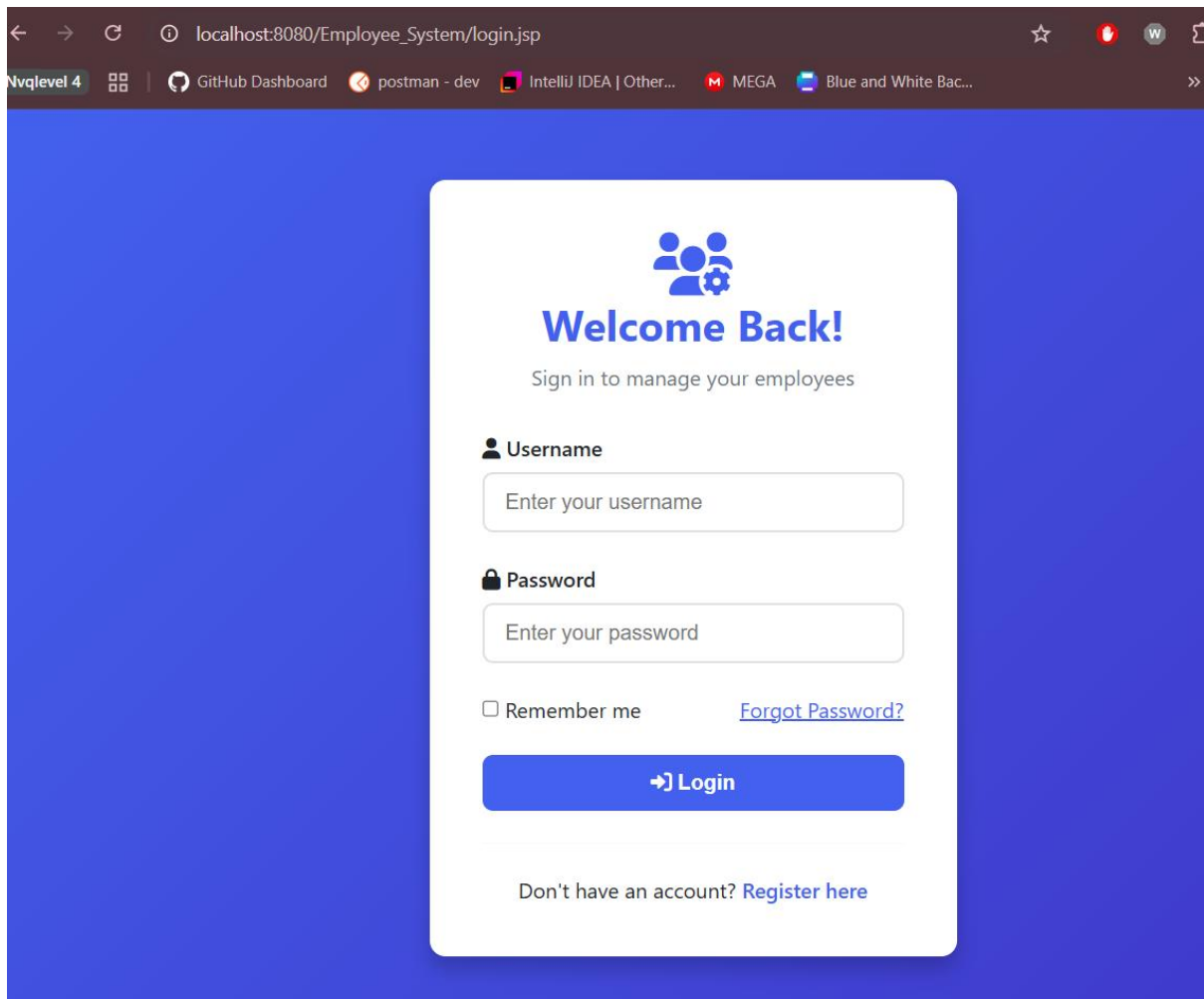
2. System analysis And Design

2.1 System Architecture



3. Frontend Development

1. login page(login.jsp)



```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Login - Employee Management System</title>
  <link rel="stylesheet" href="${pageContext.request.contextPath}/css/style.css">
  <link rel="stylesheet" href="${pageContext.request.contextPath}/css/auth.css">
  <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/font-awesome/6.0.0/css/all.min.css">
</head>
<body>
  <div class="auth-container">
    <div class="auth-card">
      <div class="auth-header">
        <i class="fas fa-users-cog" style="font-size: 3rem; color: var(--primary);"></i>
        <h1>Welcome Back!</h1>
        <p>Sign in to manage your employees</p>
      </div>
    </div>
  </div>
</body>
</html>
```

</div>

<form action="login" method="POST" id="loginForm">

<div class="form-group">

<label class="form-label" for="username">

<i class="fas fa-user"></i> Username

</label>

<input type="text" class="form-control" id="username" name="username"
placeholder="Enter your username" required>

</div>

<div class="form-group">

<label class="form-label" for="password">

<i class="fas fa-lock"></i> Password

</label>

<input type="password" class="form-control" id="password"
name="password"
placeholder="Enter your password" required>

</div>

<div class="form-group" style="display: flex; justify-content: space-between;
align-items: center;">

<label>

<input type="checkbox" name="remember"> Remember me

</label>

Forgot
Password?

</div>

<button type="submit" class="btn btn-primary" style="width: 100%;">

<i class="fas fa-sign-in-alt"></i> Login

</button>

</form>

<div class="auth-footer">

<p>Don't have an account? Register here</p>

</div>

</div>

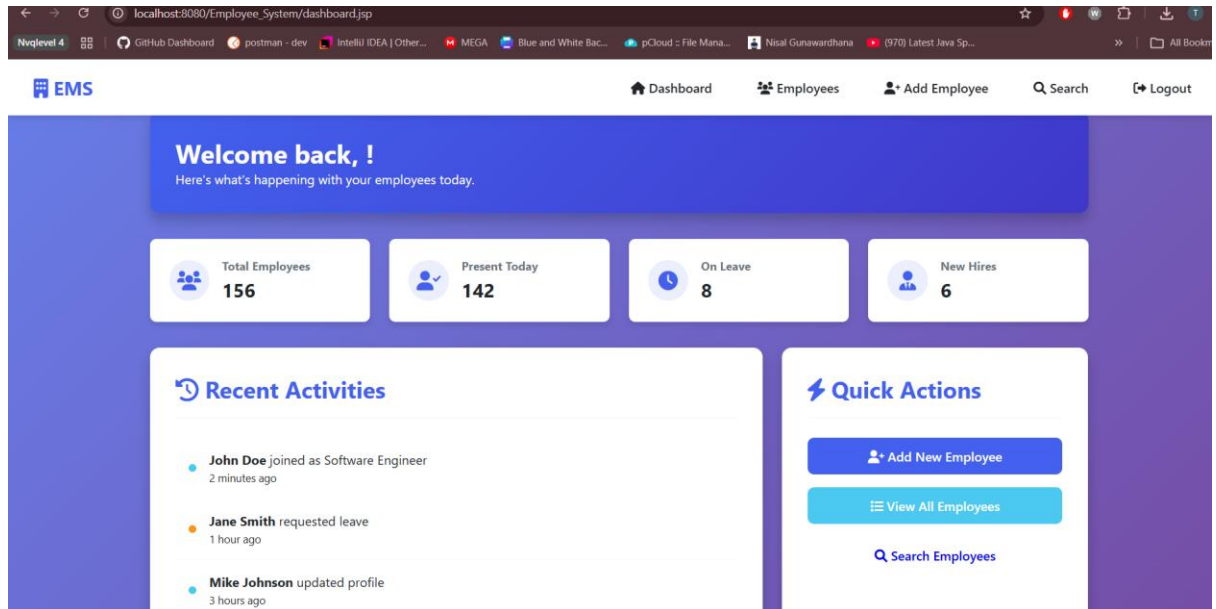
</div>

<script src="\${pageContext.request.contextPath}/js/validation.js"></script>

</body>

</html>

2. Dashboard(dashboard.jsp)



<%@ page contentType="text/html; charset=UTF-8" language="java" %>

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Dashboard - Employee Management System</title>

<link rel="stylesheet"

href="{pageContext.request.contextPath}/css/style.css">

<link rel="stylesheet"

href="{pageContext.request.contextPath}/css/dashboard.css">

<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0/css/all.min.css">

</head>

<body>

<!-- Navigation -->

<nav class="navbar">

<i class="fas fa-building"></i> EMS

<div class="navbar-nav">

<i class="fas fa-home"></i> Dashboard

```
</a>
<a href="view_employees.jsp" class="nav-link">
  <i class="fas fa-users"></i> Employees
</a>
<a href="create_employee.jsp" class="nav-link">
  <i class="fas fa-user-plus"></i> Add Employee
</a>
<a href="search_results.jsp" class="nav-link">
  <i class="fas fa-search"></i> Search
</a>
<a href="logout" class="nav-link">
  <i class="fas fa-sign-out-alt"></i> Logout
</a>
</div>
</nav>
```

```
<!-- Main Content -->
<div class="container">
  <!-- Welcome Banner -->
  <div class="card" style="background: linear-gradient(135deg, var(--primary),
var(--secondary)); color: white;">
    <h1 style="color: white;">Welcome back,
    ${sessionScope.username}</h1>
    <p>Here's what's happening with your employees today.</p>
  </div>
```

```
<!-- Statistics Cards -->
<div class="dashboard-stats">
  <div class="stat-card">
    <div class="stat-icon">
      <i class="fas fa-users"></i>
    </div>
    <div class="stat-info">
      <h3>Total Employees</h3>
      <p>156</p>
    </div>
  </div>
```

```
<div class="stat-card">
  <div class="stat-icon">
    <i class="fas fa-user-check"></i>
```

```
</div>
<div class="stat-info">
  <h3>Present Today</h3>
  <p>142</p>
</div>
</div>
```

```
<div class="stat-card">
  <div class="stat-icon">
    <i class="fas fa-clock"></i>
  </div>
  <div class="stat-info">
    <h3>On Leave</h3>
    <p>8</p>
  </div>
</div>
```

```
<div class="stat-card">
  <div class="stat-icon">
    <i class="fas fa-user-tie"></i>
  </div>
  <div class="stat-info">
    <h3>New Hires</h3>
    <p>6</p>
  </div>
</div>
</div>
```

```
<!-- Recent Activities and Quick Actions -->
<div style="display: grid; grid-template-columns: 2fr 1fr; gap: 1.5rem;">
  <!-- Recent Activities -->
  <div class="card">
    <div class="card-header">
      <h2><i class="fas fa-history"></i> Recent Activities</h2>
    </div>
    <ul class="activity-list">
      <li class="activity-item">
        <span class="activity-dot"></span>
        <div>
          <strong>John Doe</strong> joined as Software Engineer
          <br>
```

```

        <small>2 minutes ago</small>
    </div>
</li>
<li class="activity-item">
    <span class="activity-dot warning"></span>
    <div>
        <strong>Jane Smith</strong> requested leave
        <br>
        <small>1 hour ago</small>
    </div>
</li>
<li class="activity-item">
    <span class="activity-dot"></span>
    <div>
        <strong>Mike Johnson</strong> updated profile
        <br>
        <small>3 hours ago</small>
    </div>
</li>
</ul>
</div>

<!-- Quick Actions -->
<div class="card">
    <div class="card-header">
        <h2><i class="fas fa-bolt"></i> Quick Actions</h2>
    </div>
    <div style="display: flex; flex-direction: column; gap: 1rem;">
        <a href="create_employee.jsp" class="btn btn-primary">
            <i class="fas fa-user-plus"></i> Add New Employee
        </a>
        <a href="view_employees.jsp" class="btn btn-success">
            <i class="fas fa-list"></i> View All Employees
        </a>
        <a href="search_results.jsp" class="btn btn-info">
            <i class="fas fa-search"></i> Search Employees
        </a>
    </div>
</div>
</div>
</div>

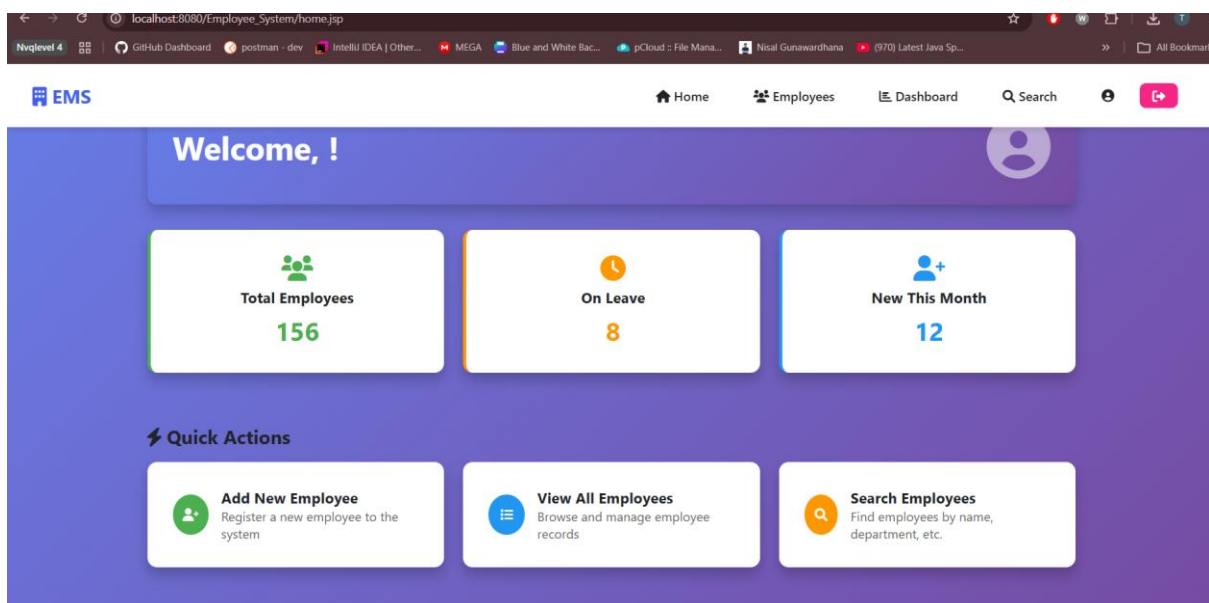
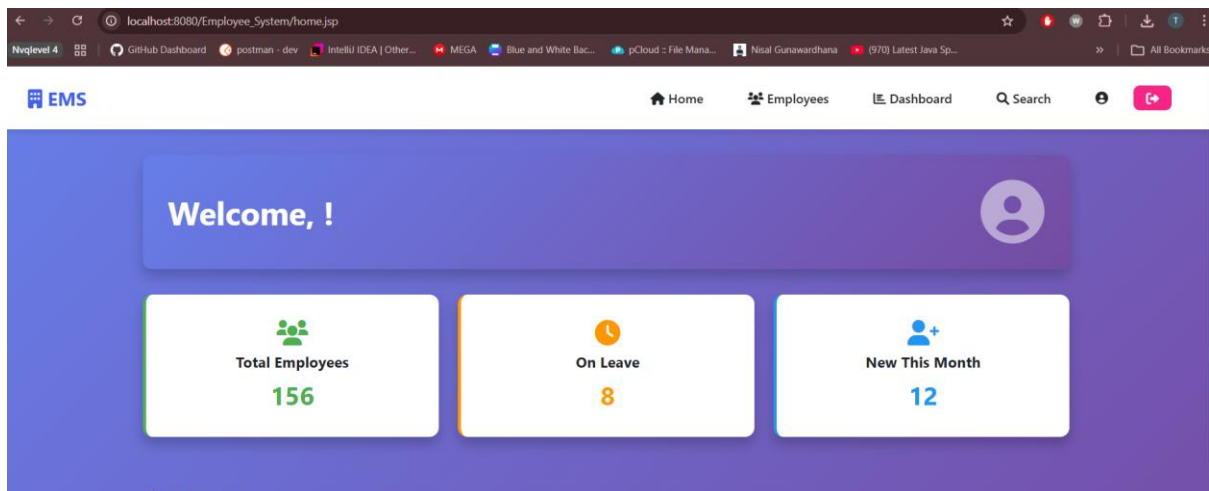
```

```

<script src="${pageContext.request.contextPath}/js/dashboard.js"></script>
</body>
</html>

```

3. Home page (home.jsp)



```

<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Dashboard - Employee Management System</title>
  <link rel="stylesheet" href="${pageContext.request.contextPath}/css/style.css">

```

```

    <link rel="stylesheet"
href="${pageContext.request.contextPath}/css/dashboard.css">
    <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.0.0/css/all.min.css">
</head>
<body>
    <!-- Navigation -->
    <nav class="navbar">
        <a href="dashboard.jsp" class="navbar-brand">
            <i class="fas fa-building"></i> EMS
        </a>
        <div class="navbar-nav">
            <a href="dashboard.jsp" class="nav-link active">
                <i class="fas fa-home"></i> Dashboard
            </a>
            <a href="view_employees.jsp" class="nav-link">
                <i class="fas fa-users"></i> Employees
            </a>
            <a href="create_employee.jsp" class="nav-link">
                <i class="fas fa-user-plus"></i> Add Employee
            </a>
            <a href="search_results.jsp" class="nav-link">
                <i class="fas fa-search"></i> Search
            </a>
            <a href="logout" class="nav-link">
                <i class="fas fa-sign-out-alt"></i> Logout
            </a>
        </div>
    </nav>

    <!-- Main Content -->
    <div class="container">
        <!-- Welcome Banner -->
        <div class="card" style="background: linear-gradient(135deg, var(--primary), var(--
secondary)); color: white;">
            <h1 style="color: white;">Welcome back, ${sessionScope.username}!</h1>
            <p>Here's what's happening with your employees today.</p>
        </div>

        <!-- Statistics Cards -->
        <div class="dashboard-stats">

```

```
<div class="stat-card">
  <div class="stat-icon">
    <i class="fas fa-users"></i>
  </div>
  <div class="stat-info">
    <h3>Total Employees</h3>
    <p>156</p>
  </div>
</div>
```

```
<div class="stat-card">
  <div class="stat-icon">
    <i class="fas fa-user-check"></i>
  </div>
  <div class="stat-info">
    <h3>Present Today</h3>
    <p>142</p>
  </div>
</div>
```

```
<div class="stat-card">
  <div class="stat-icon">
    <i class="fas fa-clock"></i>
  </div>
  <div class="stat-info">
    <h3>On Leave</h3>
    <p>8</p>
  </div>
</div>
```

```
<div class="stat-card">
  <div class="stat-icon">
    <i class="fas fa-user-tie"></i>
  </div>
  <div class="stat-info">
    <h3>New Hires</h3>
    <p>6</p>
  </div>
</div>
</div>
```

```
<!-- Recent Activities and Quick Actions -->
<div style="display: grid; grid-template-columns: 2fr 1fr; gap: 1.5rem;">
  <!-- Recent Activities -->
  <div class="card">
    <div class="card-header">
      <h2><i class="fas fa-history"></i> Recent Activities</h2>
    </div>
    <ul class="activity-list">
      <li class="activity-item">
        <span class="activity-dot"></span>
        <div>
          <strong>John Doe</strong> joined as Software Engineer
          <br>
          <small>2 minutes ago</small>
        </div>
      </li>
      <li class="activity-item">
        <span class="activity-dot warning"></span>
        <div>
          <strong>Jane Smith</strong> requested leave
          <br>
          <small>1 hour ago</small>
        </div>
      </li>
      <li class="activity-item">
        <span class="activity-dot"></span>
        <div>
          <strong>Mike Johnson</strong> updated profile
          <br>
          <small>3 hours ago</small>
        </div>
      </li>
    </ul>
  </div>

  <!-- Quick Actions -->
  <div class="card">
    <div class="card-header">
      <h2><i class="fas fa-bolt"></i> Quick Actions</h2>
    </div>
    <div style="display: flex; flex-direction: column; gap: 1rem;">
```

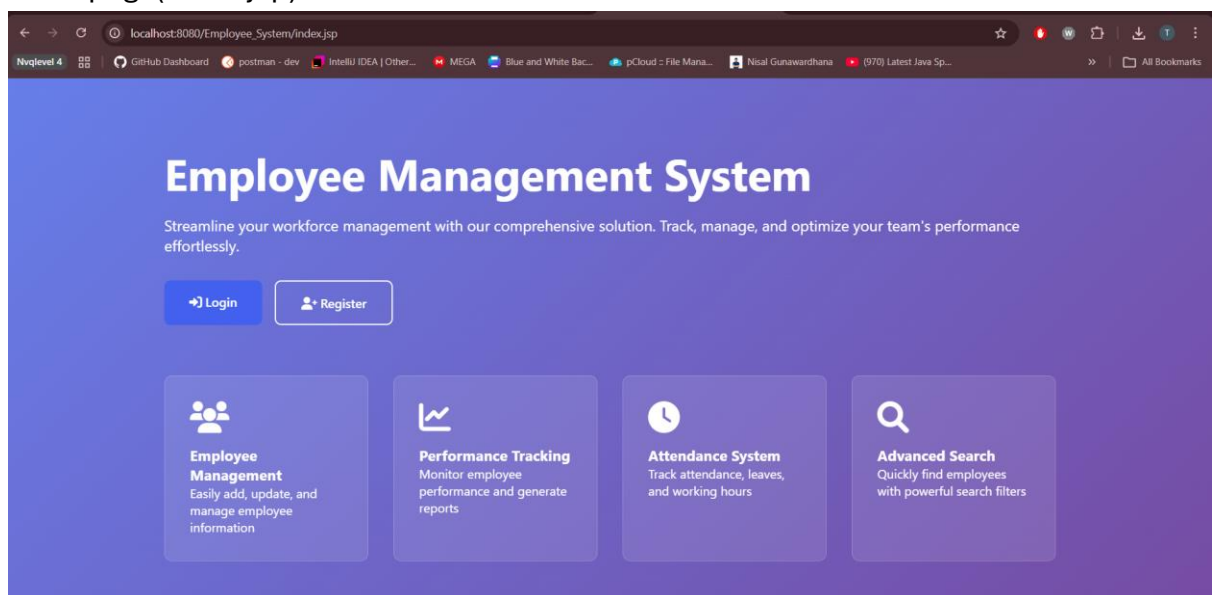
```

<a href="create_employee.jsp" class="btn btn-primary">
  <i class="fas fa-user-plus"></i> Add New Employee
</a>
<a href="view_employees.jsp" class="btn btn-success">
  <i class="fas fa-list"></i> View All Employees
</a>
<a href="search_results.jsp" class="btn btn-info">
  <i class="fas fa-search"></i> Search Employees
</a>
</div>
</div>
</div>
</div>

<script src="${pageContext.request.contextPath}/js/dashboard.js"></script>
</body>
</html>

```

4. index page(index.jsp)



```

<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Employee Management System</title>

```

```
<link rel="stylesheet"
href="{pageContext.request.contextPath}/css/style.css">
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.0.0/css/all.min.css">
<style>
    .hero-section {
        background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
        min-height: 100vh;
        display: flex;
        align-items: center;
        color: white;
    }
    .hero-content {
        max-width: 1200px;
        margin: 0 auto;
        padding: 2rem;
    }
    .hero-title {
        font-size: 3.5rem;
        margin-bottom: 1rem;
        animation: fadeInUp 1s ease;
    }
    .hero-subtitle {
        font-size: 1.2rem;
        margin-bottom: 2rem;
        opacity: 0.9;
        animation: fadeInUp 1s ease 0.2s both;
    }
    .feature-grid {
        display: grid;
        grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
        gap: 2rem;
        margin-top: 4rem;
        animation: fadeInUp 1s ease 0.4s both;
    }
    .feature-card {
        background: rgba(255,255,255,0.1);
        padding: 2rem;
        border-radius: 10px;
        backdrop-filter: blur(10px);
        border: 1px solid rgba(255,255,255,0.2);
    }
```

```

        transition: transform 0.3s;
    }
    .feature-card:hover {
        transform: translateY(-10px);
    }
    .feature-card i {
        font-size: 2.5rem;
        margin-bottom: 1rem;
    }
    @keyframes fadeInUp {
        from {
            opacity: 0;
            transform: translateY(30px);
        }
        to {
            opacity: 1;
            transform: translateY(0);
        }
    }
    .cta-buttons {
        display: flex;
        gap: 1rem;
        animation: fadeInUp 1s ease 0.3s both;
    }
    .btn-outline-light {
        border: 2px solid white;
        background: transparent;
        color: white;
    }
    .btn-outline-light:hover {
        background: white;
        color: #764ba2;
    }
</style>
</head>
<body>
    <div class="hero-section">
        <div class="hero-content">
            <h1 class="hero-title">Employee Management System</h1>
            <p class="hero-subtitle">Streamline your workforce management with our
comprehensive solution. Track, manage, and optimize your team's performance

```

effortlessly.</p>

```
<div class="cta-buttons">
  <a href="auth/login.jsp" class="btn btn-primary" style="padding: 1rem
2rem;">
    <i class="fas fa-sign-in-alt"></i> Login
  </a>
  <a href="auth/register.jsp" class="btn btn-outline-light" style="padding:
1rem 2rem;">
    <i class="fas fa-user-plus"></i> Register
  </a>
</div>
```

```
<div class="feature-grid">
  <div class="feature-card">
    <i class="fas fa-users"></i>
    <h3>Employee Management</h3>
    <p>Easily add, update, and manage employee information</p>
  </div>
  <div class="feature-card">
    <i class="fas fa-chart-line"></i>
    <h3>Performance Tracking</h3>
    <p>Monitor employee performance and generate reports</p>
  </div>
  <div class="feature-card">
    <i class="fas fa-clock"></i>
    <h3>Attendance System</h3>
    <p>Track attendance, leaves, and working hours</p>
  </div>
  <div class="feature-card">
    <i class="fas fa-search"></i>
    <h3>Advanced Search</h3>
    <p>Quickly find employees with powerful search filters</p>
  </div>
</div>
</div>
</div>
</body>
</html>
```

5. Register Page (registra.jsp)

localhost:8080/Employee_System/register.jsp

Create Account
Register to start managing employees

Account Information

Username *
Choose a username

Email Address *
Enter your email

Password *
Create password

Confirm Password *
Confirm password

localhost:8080/Employee_System/register.jsp

Personal Information

First Name *
First name

Last Name *
Last name

Phone Number
Enter phone number

Company Information

Company Name
Enter company name

Role in Company
Select Role

☐ I agree to the [Terms and Conditions](#) and [Privacy Policy](#)

Register

Already have an account? [Login here](#)

```
<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
```

```

<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Register - Employee Management System</title>
<link rel="stylesheet" href="{pageContext.request.contextPath}/css/style.css">
<link rel="stylesheet" href="{pageContext.request.contextPath}/css/auth.css">
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.0.0/css/all.min.css">
</head>
<body>
  <div class="auth-container">
    <div class="auth-card" style="max-width: 500px;">
      <div class="auth-header">
        <i class="fas fa-user-plus" style="font-size: 3rem; color: var(--primary);"></i>
        <h1>Create Account</h1>
        <p>Register to start managing employees</p>
      </div>

      <!-- Registration Form -->
      <form action="register" method="POST" id="registerForm">
        <!-- Account Information -->
        <h4 style="color: #667eea; margin-bottom: 1rem;">Account Information</h4>

        <div class="form-group">
          <label class="form-label" for="username">
            <i class="fas fa-user"></i> Username *
          </label>
          <input type="text" class="form-control" id="username" name="username"
            placeholder="Choose a username" required>
        </div>

        <div class="form-group">
          <label class="form-label" for="email">
            <i class="fas fa-envelope"></i> Email Address *
          </label>
          <input type="email" class="form-control" id="email" name="email"
            placeholder="Enter your email" required>
        </div>

        <div style="display: grid; grid-template-columns: 1fr 1fr; gap: 1rem;">
          <div class="form-group">
            <label class="form-label" for="password">
              <i class="fas fa-lock"></i> Password *

```

```

        </label>
        <input type="password" class="form-control" id="password"
name="password"
        placeholder="Create password" required>
    </div>

    <div class="form-group">
        <label class="form-label" for="confirmPassword">
            <i class="fas fa-lock"></i> Confirm Password *
        </label>
        <input type="password" class="form-control" id="confirmPassword"
            name="confirmPassword" placeholder="Confirm password" required>
        </div>
    </div>

    <!-- Personal Information -->
    <h4 style="color: #667eea; margin: 1.5rem 0 1rem;">Personal Information</h4>

    <div style="display: grid; grid-template-columns: 1fr 1fr; gap: 1rem;">
        <div class="form-group">
            <label class="form-label" for="firstName">First Name *</label>
            <input type="text" class="form-control" id="firstName" name="firstName"
                placeholder="First name" required>
        </div>

        <div class="form-group">
            <label class="form-label" for="lastName">Last Name *</label>
            <input type="text" class="form-control" id="lastName" name="lastName"
                placeholder="Last name" required>
        </div>
    </div>

    <div class="form-group">
        <label class="form-label" for="phone">Phone Number</label>
        <input type="tel" class="form-control" id="phone" name="phone"
            placeholder="Enter phone number">
    </div>

    <!-- Company Information -->
    <h4 style="color: #667eea; margin: 1.5rem 0 1rem;">Company
Information</h4>

```

```

<div class="form-group">
  <label class="form-label" for="company">Company Name</label>
  <input type="text" class="form-control" id="company" name="company"
    placeholder="Enter company name">
</div>

```

```

<div class="form-group">
  <label class="form-label" for="role">Role in Company</label>
  <select class="form-control" id="role" name="role">
    <option value="">Select Role</option>
    <option value="admin">Administrator</option>
    <option value="manager">Manager</option>
    <option value="hr">HR Staff</option>
    <option value="employee">Employee</option>
  </select>
</div>

```

```

<!-- Terms and Conditions -->
<div class="form-group" style="margin: 1.5rem 0;">
  <label style="display: flex; align-items: center; gap: 0.5rem;">
    <input type="checkbox" name="terms" required>
    I agree to the <a href="#" style="color: #667eea;">Terms and Conditions</a>
    and
    <a href="#" style="color: #667eea;">Privacy Policy</a>
  </label>
</div>

```

```

<!-- Password Strength Indicator -->
<div id="passwordStrength" style="margin-bottom: 1rem; display: none;">
  <div style="display: flex; gap: 0.5rem; margin-bottom: 0.5rem;">
    <div style="flex: 1; height: 5px; background: #ddd; border-radius: 5px;">
      <div id="strengthBar" style="height: 100%; width: 0%; background:
#dc3545; border-radius: 5px;"></div>
    </div>
  </div>
  <p id="strengthText" style="font-size: 0.9rem; color: #666;">Password
strength: Weak</p>
</div>

```

```

<button type="submit" class="btn btn-primary" style="width: 100%; margin-top:

```

```

1rem;">
    <i class="fas fa-user-plus"></i> Register
  </button>
</form>

<div class="auth-footer">
  <p>Already have an account? <a href="login.jsp">Login here</a></p>
</div>
</div>
</div>

<script>
  // Password strength checker
  document.getElementById('password').addEventListener('input', function() {
    const password = this.value;
    const strengthDiv = document.getElementById('passwordStrength');
    const strengthBar = document.getElementById('strengthBar');
    const strengthText = document.getElementById('strengthText');

    if(password.length > 0) {
      strengthDiv.style.display = 'block';

      let strength = 0;
      let color = '#dc3545';
      let text = 'Weak';

      if(password.length >= 8) strength++;
      if(password.match(/[a-z]+/)) strength++;
      if(password.match(/[A-Z]+/)) strength++;
      if(password.match(/[0-9]+/)) strength++;
      if(password.match(/[$@#&!]+/)) strength++;

      switch(strength) {
        case 0:
        case 1:
        case 2:
          color = '#dc3545';
          text = 'Weak';
          break;
        case 3:
          color = '#ffc107';

```

```

        text = 'Medium';
        break;
    case 4:
        color = '#28a745';
        text = 'Strong';
        break;
    case 5:
        color = '#28a745';
        text = 'Very Strong';
        break;
    }

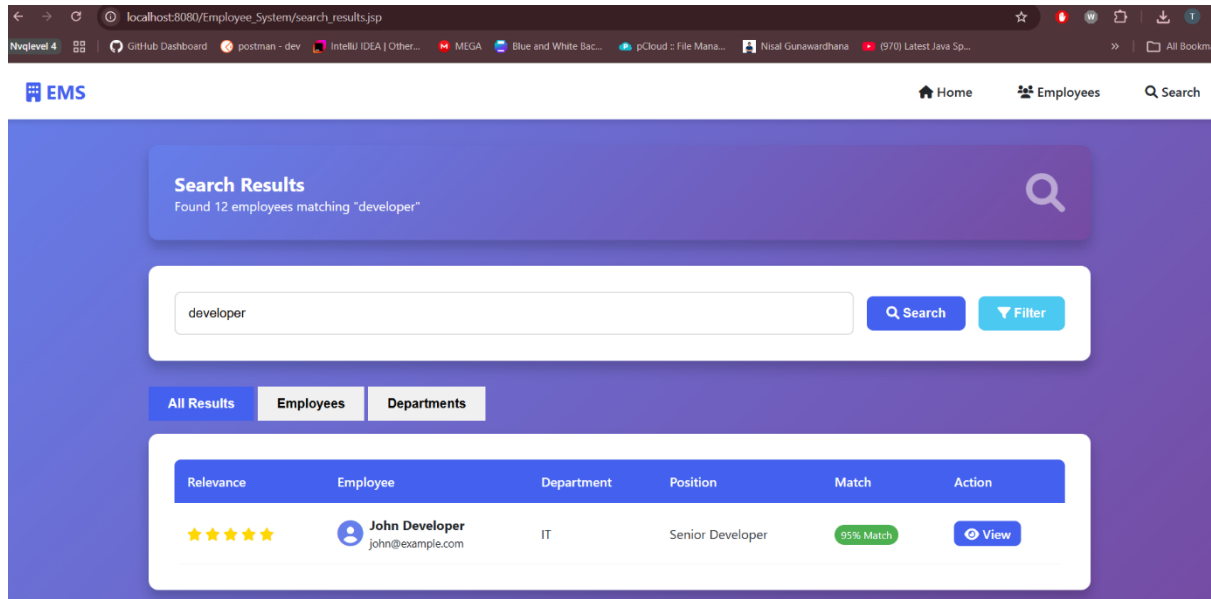
    strengthBar.style.width = (strength * 20) + '%';
    strengthBar.style.background = color;
    strengthText.innerHTML = 'Password strength: ' + text;
} else {
    strengthDiv.style.display = 'none';
}
});

// Form validation
document.getElementById('registerForm').addEventListener('submit', function(e) {
    const password = document.getElementById('password').value;
    const confirmPassword = document.getElementById('confirmPassword').value;

    if(password !== confirmPassword) {
        e.preventDefault();
        alert('Passwords do not match!');
    }
});
</script>
</body>
</html>

```

6. Research page(research_result.jsp)



```
<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Search Results</title>
  <link rel="stylesheet" href="${pageContext.request.contextPath}/css/style.css">
  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.0.0/css/all.min.css">
</head>
<body>
  <nav class="navbar">
    <a href="../home.jsp" class="navbar-brand">
      <i class="fas fa-building"></i> EMS
    </a>
    <div class="navbar-nav">
      <a href="../home.jsp" class="nav-link">
        <i class="fas fa-home"></i> Home
      </a>
      <a href="view_employees.jsp" class="nav-link">
        <i class="fas fa-users"></i> Employees
      </a>
      <a href="search_results.jsp" class="nav-link active">
```

```
        <i class="fas fa-search"></i> Search
    </a>
</div>
</nav>
```

```
<div class="container">
    <!-- Search Summary -->
    <div class="card" style="background: linear-gradient(135deg, #667eea, #764ba2);
color: white;">
        <div style="display: flex; justify-content: space-between; align-items: center;">
            <div>
                <h2 style="color: white;">Search Results</h2>
                <p>Found 12 employees matching "developer"</p>
            </div>
            <i class="fas fa-search" style="font-size: 3rem; opacity: 0.5;"></i>
        </div>
    </div>
```

```
    <!-- Search Filters -->
    <div class="card">
        <form action="search_results.jsp" method="GET" class="search-box">
            <input type="text" name="q" class="search-input" value="developer"
placeholder="Search...">
            <button type="submit" class="btn btn-primary">
                <i class="fas fa-search"></i> Search
            </button>
            <button type="button" class="btn btn-success" onclick="filterResults()">
                <i class="fas fa-filter"></i> Filter
            </button>
        </form>
    </div>
```

```
    <!-- Results Tabs -->
    <div style="margin-bottom: 1rem;">
        <button class="btn btn-primary" style="border-radius: 0;"
onclick="showTab('all')">All Results</button>
        <button class="btn" style="border-radius: 0;"
onclick="showTab('employees')">Employees</button>
        <button class="btn" style="border-radius: 0;"
onclick="showTab('departments')">Departments</button>
    </div>
```

```

<!-- Search Results Table -->
<div class="card">
  <div class="table-container">
    <table>
      <thead>
        <tr>
          <th>Relevance</th>
          <th>Employee</th>
          <th>Department</th>
          <th>Position</th>
          <th>Match</th>
          <th>Action</th>
        </tr>
      </thead>
      <tbody>
        <!-- Sample Results -->
        <tr>
          <td>
            <i class="fas fa-star" style="color: #FFD700;"></i>
            <i class="fas fa-star" style="color: #FFD700;"></i>
            <i class="fas fa-star" style="color: #FFD700;"></i>
            <i class="fas fa-star" style="color: #FFD700;"></i>
            <i class="fas fa-star" style="color: #FFD700;"></i>
          </td>
          <td>
            <div style="display: flex; align-items: center; gap: 0.5rem;">
              <i class="fas fa-user-circle" style="font-size: 2rem; color:
#667eea;"></i>
              <div>
                <strong>John Developer</strong>
                <br>
                <small>john@example.com</small>
              </div>
            </div>
          </td>
          <td>IT</td>
          <td>Senior Developer</td>
          <td>
            <span style="background: #4CAF50; color: white; padding: 0.25rem
0.5rem; border-radius: 20px; font-size: 0.8rem;">

```

```

                95% Match
            </span>
        </td>
        <td>
            <a href="view_employees.jsp" class="btn btn-primary" style="padding:
0.3rem 0.8rem;">
                <i class="fas fa-eye"></i> View
            </a>
        </td>
    </tr>
    <!-- More results... -->
</tbody>
</table>
</div>
</div>

```

```

<!-- No Results Message (Hidden by default) -->
<div id="noResults" class="card" style="text-align: center; display: none;">
    <i class="fas fa-frown" style="font-size: 4rem; color: #6c757d; margin-bottom:
1rem;"></i>
    <h3>No results found</h3>
    <p>Try adjusting your search or filters</p>
    <a href="view_employees.jsp" class="btn btn-primary">Browse All
Employees</a>
</div>
</div>

```

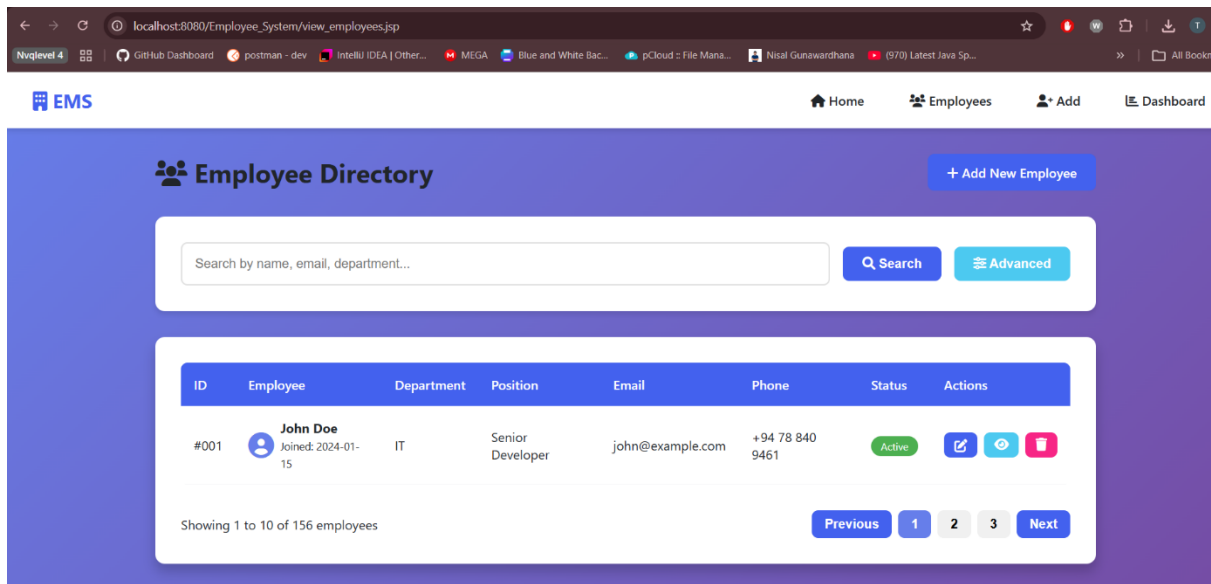
```

<script>
    function filterResults() {
        // Implement filter logic
        alert('Filter options will appear here');
    }

    function showTab(tabName) {
        // Implement tab switching
        console.log('Showing tab:', tabName);
    }
</script>
</body>
</html>

```

7. Update _Employee_Page(update_Employee.jsp)



```
<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Update Employee</title>
  <link rel="stylesheet" href="${pageContext.request.contextPath}/css/style.css">
  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.0.0/css/all.min.css">
</head>
<body>
  <nav class="navbar">
    <a href="../home.jsp" class="navbar-brand">
      <i class="fas fa-building"></i> EMS
    </a>
    <div class="navbar-nav">
      <a href="../home.jsp" class="nav-link">
        <i class="fas fa-home"></i> Home
      </a>
      <a href="view_employees.jsp" class="nav-link">
        <i class="fas fa-users"></i> Employees
      </a>
    </div>
  </nav>
```

```
</nav>
```

```
<div class="container">
```

```
  <div class="card" style="max-width: 800px; margin: 0 auto;">
```

```
    <div class="card-header">
```

```
      <h2><i class="fas fa-edit"></i> Update Employee</h2>
```

```
      <p>Edit employee information - ID: ${param.id}</p>
```

```
    </div>
```

```
    <form action="update" method="POST" id="updateForm">
```

```
      <input type="hidden" name="id" value="${param.id}">
```

```
      <!-- Personal Information -->
```

```
      <h3 style="color: #667eea; margin: 1.5rem 0 1rem;">Personal Information</h3>
```

```
      <div style="display: grid; grid-template-columns: repeat(2, 1fr); gap: 1rem;">
```

```
        <div class="form-group">
```

```
          <label class="form-label">First Name</label>
```

```
          <input type="text" class="form-control" name="firstName" value="John">
```

```
        </div>
```

```
        <div class="form-group">
```

```
          <label class="form-label">Last Name</label>
```

```
          <input type="text" class="form-control" name="lastName" value="Doe">
```

```
        </div>
```

```
      </div>
```

```
      <div class="form-group">
```

```
        <label class="form-label">Email Address</label>
```

```
        <input type="email" class="form-control" name="email" value="john.doe@example.com">
```

```
      </div>
```

```
      <div style="display: grid; grid-template-columns: repeat(2, 1fr); gap: 1rem;">
```

```
        <div class="form-group">
```

```
          <label class="form-label">Phone Number</label>
```

```
          <input type="tel" class="form-control" name="phone" value="+1 234 567 890">
```

```
        </div>
```

```
      <div class="form-group">
```

```

        <label class="form-label">Department</label>
        <select class="form-control" name="department">
            <option value="it" selected>IT</option>
            <option value="hr">HR</option>
            <option value="finance">Finance</option>
        </select>
    </div>
</div>

<div style="display: grid; grid-template-columns: repeat(2, 1fr); gap: 1rem;">
    <div class="form-group">
        <label class="form-label">Position</label>
        <input type="text" class="form-control" name="position" value="Senior
Developer">
    </div>

    <div class="form-group">
        <label class="form-label">Salary (LKR)</label>
        <input type="number" class="form-control" name="salary"
value="150000">
    </div>
</div>

<div class="form-group">
    <label class="form-label">Status</label>
    <select class="form-control" name="status">
        <option value="active" selected>Active</option>
        <option value="inactive">Inactive</option>
        <option value="onleave">On Leave</option>
    </select>
</div>

<!-- Form Actions -->
<div style="display: flex; gap: 1rem; margin-top: 2rem;">
    <button type="submit" class="btn btn-primary" style="flex: 1;">
        <i class="fas fa-save"></i> Update Employee
    </button>
    <a href="view_employees.jsp" class="btn" style="flex: 1; background:
#6c757d; color: white;">
        <i class="fas fa-arrow-left"></i> Back
    </a>

```

```

        </div>
    </form>
</div>
</div>

<script>
    // Confirmation before update
    document.getElementById('updateForm').addEventListener('submit', function(e) {
        if(!confirm('Are you sure you want to update this employee?')) {
            e.preventDefault();
        }
    });
</script>
</body>
</html>

```

8. View_Employee Page(view_employees.jsp)

EMS Home Employees

Update Employee

Edit employee information - ID:

Personal Information

First Name	Last Name
John	Doe
Email Address	
john.doe@example.com	
Phone Number	Department
+1 234 567 890	IT
Position	Salary (LKR)
Senior Developer	150000
Status	
Active	

```
<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>View Employees - Employee Management</title>
  <link rel="stylesheet" href="${pageContext.request.contextPath}/css/style.css">
  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.0.0/css/all.min.css">
</head>
<body>
  <nav class="navbar">
    <a href="../home.jsp" class="navbar-brand">
      <i class="fas fa-building"></i> EMS
    </a>
    <div class="navbar-nav">
      <a href="../home.jsp" class="nav-link">
        <i class="fas fa-home"></i> Home
      </a>
      <a href="view_employees.jsp" class="nav-link active">
        <i class="fas fa-users"></i> Employees
      </a>
      <a href="create_employee.jsp" class="nav-link">
        <i class="fas fa-user-plus"></i> Add
      </a>
      <a href="dashboard.jsp" class="nav-link">
        <i class="fas fa-chart-bar"></i> Dashboard
      </a>
    </div>
  </nav>

  <div class="container">
    <!-- Header with Search and Add Button -->
    <div style="display: flex; justify-content: space-between; align-items: center;
margin-bottom: 2rem;">
      <h1><i class="fas fa-users"></i> Employee Directory</h1>
      <a href="create_employee.jsp" class="btn btn-primary">
        <i class="fas fa-plus"></i> Add New Employee
      </a>
    </div>
  </div>
</body>
</html>
```

</div>

<!-- Search Bar -->

<div class="search-container">

<form action="search_results.jsp" method="GET" class="search-box">

<input type="text" name="query" class="search-input"

placeholder="Search by name, email, department...">

<button type="submit" class="btn btn-primary">

<i class="fas fa-search"></i> Search

</button>

<button type="button" class="btn btn-success"

onclick="showAdvancedSearch()">

<i class="fas fa-sliders-h"></i> Advanced

</button>

</form>

<!-- Advanced Search Panel (Hidden by default) -->

<div id="advancedSearch" style="display: none; margin-top: 1rem;">

<div class="card">

<h3>Advanced Search</h3>

<div style="display: grid; grid-template-columns: repeat(auto-fit, minmax(200px, 1fr)); gap: 1rem;">

<div class="form-group">

<label>Department</label>

<select class="form-control">

<option>All Departments</option>

<option>IT</option>

<option>HR</option>

<option>Finance</option>

<option>Marketing</option>

</select>

</div>

<div class="form-group">

<label>Position</label>

<select class="form-control">

<option>All Positions</option>

<option>Manager</option>

<option>Developer</option>

<option>Designer</option>

</select>

</div>

```
<div class="form-group">
  <label>Status</label>
  <select class="form-control">
    <option>All</option>
    <option>Active</option>
    <option>On Leave</option>
  </select>
</div>
</div>
</div>
</div>
</div>
```

```
<!-- Employees Table -->
```

```
<div class="card">
  <div class="table-container">
    <table>
      <thead>
        <tr>
          <th>ID</th>
          <th>Employee</th>
          <th>Department</th>
          <th>Position</th>
          <th>Email</th>
          <th>Phone</th>
          <th>Status</th>
          <th>Actions</th>
        </tr>
      </thead>
      <tbody>
        <!-- Sample Data - Replace with JSTL foreach -->
        <tr>
          <td>#001</td>
          <td>
            <div style="display: flex; align-items: center; gap: 0.5rem;">
              <i class="fas fa-user-circle" style="font-size: 2rem; color:
#667eea;"></i>
              <div>
                <strong>John Doe</strong>
                <br>
                <small>Joined: 2024-01-15</small>
              </div>
            </div>
          </td>
        </tr>
      </tbody>
    </table>
  </div>
</div>
```

```

        </div>
    </div>
</td>
<td>IT</td>
<td>Senior Developer</td>
<td>john@example.com</td>
<td>+94 78 840 9461</td>
<td>
    <span style="background: #4CAF50; color: white; padding: 0.25rem
0.75rem; border-radius: 20px; font-size: 0.8rem;">
        Active
    </span>
</td>
<td>
    <div style="display: flex; gap: 0.5rem;">
        <a href="update_employee.jsp?id=1" class="btn btn-primary"
style="padding: 0.3rem 0.8rem;">
            <i class="fas fa-edit"></i>
        </a>
        <button onclick="viewEmployee('1')" class="btn btn-success"
style="padding: 0.3rem 0.8rem;">
            <i class="fas fa-eye"></i>
        </button>
        <button onclick="deleteEmployee('1')" class="btn btn-danger"
style="padding: 0.3rem 0.8rem;">
            <i class="fas fa-trash"></i>
        </button>
    </div>
</td>
</tr>
<!-- More rows as needed -->
</tbody>
</table>
</div>

<!-- Pagination -->
<div style="display: flex; justify-content: space-between; align-items: center;
margin-top: 2rem;">
    <p>Showing 1 to 10 of 156 employees</p>
    <div style="display: flex; gap: 0.5rem;">
        <button class="btn btn-primary" style="padding: 0.5rem 1rem;"

```

```

disabled>Previous</button>
    <button class="btn" style="background: #667eea; color: white; padding:
0.5rem 1rem;">1</button>
    <button class="btn" style="padding: 0.5rem 1rem;">2</button>
    <button class="btn" style="padding: 0.5rem 1rem;">3</button>
    <button class="btn btn-primary" style="padding: 0.5rem
1rem;">Next</button>
    </div>
</div>
</div>
</div>

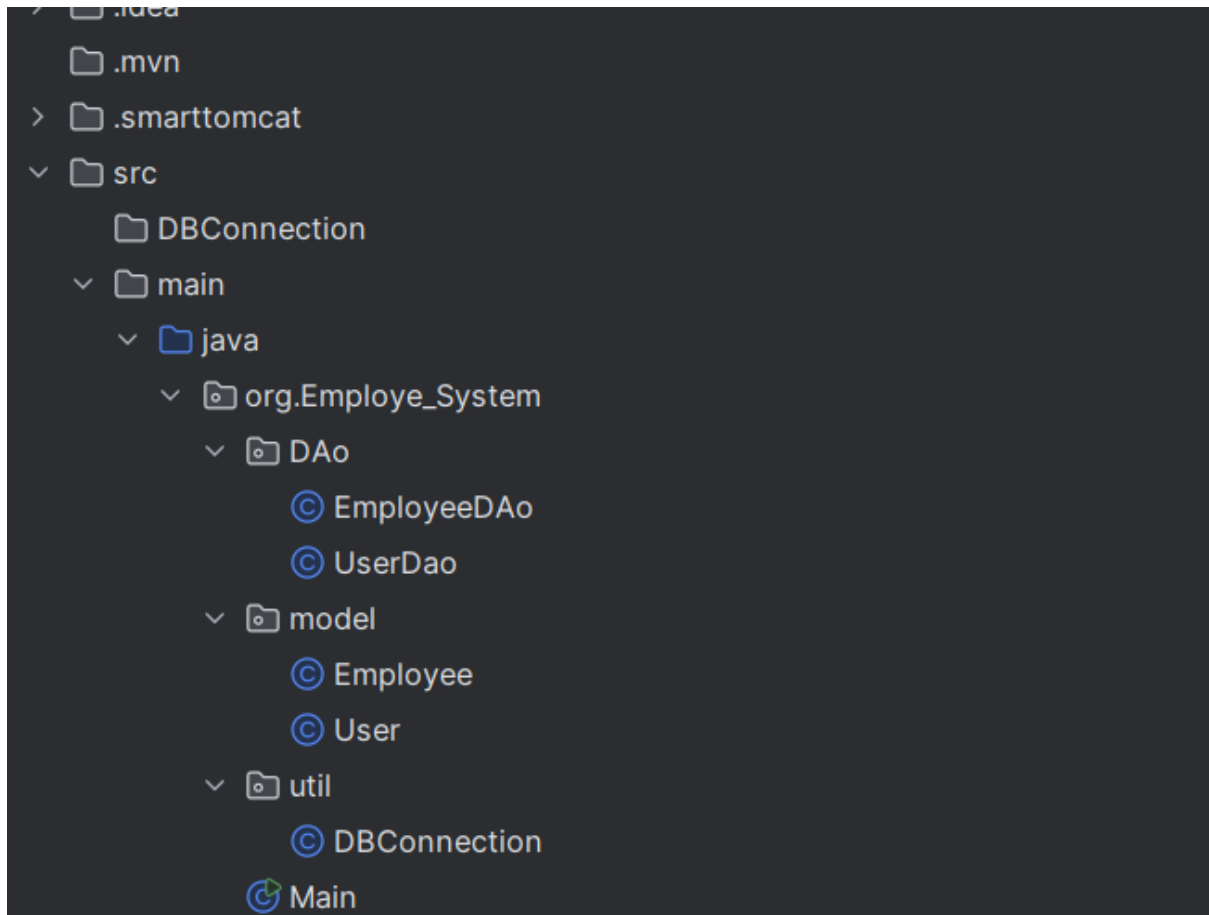
<script>
function showAdvancedSearch() {
    const panel = document.getElementById('advancedSearch');
    panel.style.display = panel.style.display === 'none' ? 'block' : 'none';
}

function viewEmployee(id) {
    window.location.href = 'view_employee.jsp?id=' + id;
}

function deleteEmployee(id) {
    if(confirm('Are you sure you want to delete this employee?')) {
        // Add delete logic here
        alert('Employee deleted successfully');
    }
}
</script>
</body>
</html>

```

4. BackEnd Development



4.1 EmployeeDao.jsp

```
package org.Employee_System.DAo;
```

```
import org.Employee_System.model.Employee;  
import org.Employee_System.util.DBConnection;
```

```
import java.sql.*;  
    import java.util.ArrayList;  
import java.util.List;
```

```
public class EmployeeDAo {
```

```
    // Create Employee
```

```
    public boolean createEmployee(Employee employee) {
```

```
        String sql = "INSERT INTO employees (first_name, last_name, dob, gender, email,  
phone, " +
```

```
        "alt_phone, address, employee_id, department, position, join_date, salary, " +
```

```
        "employment_type, photo_path, status) VALUES  
(?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?)";
```

```
try (Connection conn = DBConnection.getConnection();  
    PreparedStatement pstmt = conn.prepareStatement(sql)) {  
  
    pstmt.setString(1, employee.getFirstName());  
    pstmt.setString(2, employee.getLastName());  
    pstmt.setDate(3, employee.getDob());  
    pstmt.setString(4, employee.getGender());  
    pstmt.setString(5, employee.getEmail());  
    pstmt.setString(6, employee.getPhone());  
    pstmt.setString(7, employee.getAltPhone());  
    pstmt.setString(8, employee.getAddress());  
    pstmt.setString(9, employee.getEmployeeId());  
    pstmt.setString(10, employee.getDepartment());  
    pstmt.setString(11, employee.getPosition());  
    pstmt.setDate(12, employee.getJoinDate());  
    pstmt.setBigDecimal(13, employee.getSalary());  
    pstmt.setString(14, employee.getEmploymentType());  
    pstmt.setString(15, employee.getPhotoPath());  
    pstmt.setString(16, "active");  
  
    return pstmt.executeUpdate() > 0;  
  
} catch (SQLException e) {  
    e.printStackTrace();  
    return false;  
}  
}
```

```
// Get All Employees
```

```
public List<Employee> getAllEmployees() {  
    List<Employee> employees = new ArrayList<>();  
    String sql = "SELECT * FROM employees ORDER BY id DESC";  
  
    try (Connection conn = DBConnection.getConnection();  
        PreparedStatement pstmt = conn.prepareStatement(sql);  
        ResultSet rs = pstmt.executeQuery()) {  
  
        while (rs.next()) {
```

```

        employees.add(extractEmployeeFromResultSet(rs));
    }

    } catch (SQLException e) {
        e.printStackTrace();
    }
    return employees;
}

// Get Employee by ID
public Employee getEmployeeById(int id) {
    String sql = "SELECT * FROM employees WHERE id = ?";

    try (Connection conn = DBConnection.getConnection();
        PreparedStatement pstmt = conn.prepareStatement(sql)) {

        pstmt.setInt(1, id);
        ResultSet rs = pstmt.executeQuery();

        if (rs.next()) {
            return extractEmployeeFromResultSet(rs);
        }

    } catch (SQLException e) {
        e.printStackTrace();
    }
    return null;
}

// Update Employee
public boolean updateEmployee(Employee employee) {
    String sql = "UPDATE employees SET first_name=?, last_name=?, dob=?, gender=?,
email=?, " +
        "phone=?, alt_phone=?, address=?, department=?, position=?, join_date=?, " +
        "salary=?, employment_type=?, status=? WHERE id=?";

    try (Connection conn = DBConnection.getConnection();
        PreparedStatement pstmt = conn.prepareStatement(sql)) {

        pstmt.setString(1, employee.getFirstName());
        pstmt.setString(2, employee.getLastName());

```

```

        pstmt.setDate(3, employee.getDob());
        pstmt.setString(4, employee.getGender());
        pstmt.setString(5, employee.getEmail());
        pstmt.setString(6, employee.getPhone());
        pstmt.setString(7, employee.getAltPhone());
        pstmt.setString(8, employee.getAddress());
        pstmt.setString(9, employee.getDepartment());
        pstmt.setString(10, employee.getPosition());
        pstmt.setDate(11, employee.getJoinDate());
        pstmt.setBigDecimal(12, employee.getSalary());
        pstmt.setString(13, employee.getEmploymentType());
        pstmt.setString(14, employee.getStatus());
        pstmt.setInt(15, employee.getId());

        return pstmt.executeUpdate() > 0;

    } catch (SQLException e) {
        e.printStackTrace();
        return false;
    }
}

// Delete Employee
public boolean deleteEmployee(int id) {
    String sql = "DELETE FROM employees WHERE id = ?";

    try (Connection conn = DBConnection.getConnection();
        PreparedStatement pstmt = conn.prepareStatement(sql)) {

        pstmt.setInt(1, id);
        return pstmt.executeUpdate() > 0;

    } catch (SQLException e) {
        e.printStackTrace();
        return false;
    }
}

// Search Employees
public List<Employee> searchEmployees(String keyword) {
    List<Employee> employees = new ArrayList<>();

```

```
String sql = "SELECT * FROM employees WHERE first_name LIKE ? OR last_name  
LIKE ? " +
```

```
"OR email LIKE ? OR department LIKE ? OR position LIKE ?";
```

```
try (Connection conn = DBConnection.getConnection();  
    PreparedStatement pstmt = conn.prepareStatement(sql)) {
```

```
    String searchPattern = "%" + keyword + "%";
```

```
    pstmt.setString(1, searchPattern);
```

```
    pstmt.setString(2, searchPattern);
```

```
    pstmt.setString(3, searchPattern);
```

```
    pstmt.setString(4, searchPattern);
```

```
    pstmt.setString(5, searchPattern);
```

```
    ResultSet rs = pstmt.executeQuery();
```

```
    while (rs.next()) {
```

```
        employees.add(extractEmployeeFromResultSet(rs));
```

```
    }
```

```
} catch (SQLException e) {
```

```
    e.printStackTrace();
```

```
}
```

```
return employees;
```

```
}
```

```
// Get Dashboard Statistics
```

```
public int getTotalEmployees() {
```

```
    String sql = "SELECT COUNT(*) FROM employees";
```

```
    try (Connection conn = DBConnection.getConnection();
```

```
        PreparedStatement pstmt = conn.prepareStatement(sql);
```

```
        ResultSet rs = pstmt.executeQuery()) {
```

```
        if (rs.next()) return rs.getInt(1);
```

```
    } catch (SQLException e) {
```

```
        e.printStackTrace();
```

```
    }
```

```
    return 0;
```

```
}
```

```
public int getActiveEmployees() {
```

```
    String sql = "SELECT COUNT(*) FROM employees WHERE status = 'active'";
```

```

try (Connection conn = DBConnection.getConnection();
    PreparedStatement pstmt = conn.prepareStatement(sql);
    ResultSet rs = pstmt.executeQuery()) {
    if (rs.next()) return rs.getInt(1);
} catch (SQLException e) {
    e.printStackTrace();
}
return 0;
}

```

```

public int getOnLeaveEmployees() {
    String sql = "SELECT COUNT(*) FROM employees WHERE status = 'onleave'";
    try (Connection conn = DBConnection.getConnection();
        PreparedStatement pstmt = conn.prepareStatement(sql);
        ResultSet rs = pstmt.executeQuery()) {
        if (rs.next()) return rs.getInt(1);
    } catch (SQLException e) {
        e.printStackTrace();
    }
    return 0;
}

```

```

public int getNewHiresThisMonth() {
    String sql = "SELECT COUNT(*) FROM employees WHERE MONTH(join_date) =
MONTH(CURRENT_DATE()) " +
        "AND YEAR(join_date) = YEAR(CURRENT_DATE())";
    try (Connection conn = DBConnection.getConnection();
        PreparedStatement pstmt = conn.prepareStatement(sql);
        ResultSet rs = pstmt.executeQuery()) {
        if (rs.next()) return rs.getInt(1);
    } catch (SQLException e) {
        e.printStackTrace();
    }
    return 0;
}

```

```

private Employee extractEmployeeFromResultSet(ResultSet rs) throws SQLException
{
    Employee emp = new Employee();
    emp.setId(rs.getInt("id"));
    emp.setFirstName(rs.getString("first_name"));
}

```

```

emp.setLastName(rs.getString("last_name"));
emp.setDob(rs.getDate("dob"));
emp.setGender(rs.getString("gender"));
emp.setEmail(rs.getString("email"));
emp.setPhone(rs.getString("phone"));
emp.setAltPhone(rs.getString("alt_phone"));
emp.setAddress(rs.getString("address"));
emp.setEmployeeId(rs.getString("employee_id"));
emp.setDepartment(rs.getString("department"));
emp.setPosition(rs.getString("position"));
emp.setJoinDate(rs.getDate("join_date"));
emp.setSalary(rs.getBigDecimal("salary"));
emp.setEmploymentType(rs.getString("employment_type"));
emp.setPhotoPath(rs.getString("photo_path"));
emp.setStatus(rs.getString("status"));
emp.setCreatedAt(rs.getDate("created_at"));
emp.setUpdatedAt(rs.getDate("updated_at"));
return emp;
}
}

```

4.2 userDao.java

```

package org.Employe_System.DAo;

import org.Employe_System.model.User;
import org.Employe_System.util.DBConnection;

import java.sql.*;

public class UserDao {

    // Register User
    public boolean registerUser(User user) {
        String sql = "INSERT INTO users (username, password, email,
first_name, last_name, phone, company, role) " +
            "VALUES (?, ?, ?, ?, ?, ?, ?, ?)";
    }
}

```

```

try (Connection conn = DBConnection.getConnection();
    PreparedStatement pstmt = conn.prepareStatement(sql)) {

    pstmt.setString(1, user.getUsername());
    pstmt.setString(2, user.getPassword()); // Hash this in real
application
    pstmt.setString(3, user.getEmail());
    pstmt.setString(4, user.getFirstName());
    pstmt.setString(5, user.getLastName());
    pstmt.setString(6, user.getPhone());
    pstmt.setString(7, user.getCompany());
    pstmt.setString(8, "USER"); // Default role

    return pstmt.executeUpdate() > 0;

} catch (SQLException e) {
    e.printStackTrace();
    return false;
}
}

// Login User
public User loginUser(String username, String password) {
    String sql = "SELECT * FROM users WHERE username = ? AND
password = ?";

    try (Connection conn = DBConnection.getConnection();
        PreparedStatement pstmt = conn.prepareStatement(sql)) {

        pstmt.setString(1, username);
        pstmt.setString(2, password); // Compare with hashed password in
real app

        ResultSet rs = pstmt.executeQuery();

        if (rs.next()) {
            User user = new User();
            user.setId(rs.getInt("id"));
            user.setUsername(rs.getString("username"));
            user.setPassword(rs.getString("password"));
            user.setEmail(rs.getString("email"));

```

```

        user.setFirstName(rs.getString("first_name"));
        user.setLastName(rs.getString("last_name"));
        user.setPhone(rs.getString("phone"));
        user.setCompany(rs.getString("company"));
        user.setRole(rs.getString("role"));
        user.setCreatedAt(rs.getTimestamp("created_at"));
        return user;
    }

    } catch (SQLException e) {
        e.printStackTrace();
    }
    return null;
}

// Check if username exists
public boolean isUsernameExists(String username) {
    String sql = "SELECT id FROM users WHERE username = ?";

    try (Connection conn = DBConnection.getConnection();
        PreparedStatement pstmt = conn.prepareStatement(sql)) {

        pstmt.setString(1, username);
        ResultSet rs = pstmt.executeQuery();
        return rs.next();

    } catch (SQLException e) {
        e.printStackTrace();
        return false;
    }
}
}

```

Database:-

UnnamedDatabase: employee_dbTable: employeesDataEmployee.sql*

BasicOptionsIndexes (1)Foreign keys (0)Check constraints (0)PartitionsCREATE codeALTER code

Name:employees

Comment:

columns: AddRemoveUpDown

#	Name	Datatype	Length/Set	Unsigned	Allow N...	Zerofill	Default	Comment	Collation	Expression	Virtuality	SRID	Invisib...
1	id	INT		☐	☐	☐	AUTO_INCREME...						☐
2	name	VARCHAR	100	☐	☐	☐	No default		utf8mb4_0900_ai_ci				☐
3	position	VARCHAR	100	☐	☐	☐	No default		utf8mb4_0900_ai_ci				☐
4	department	VARCHAR	100	☐	☐	☐	No default		utf8mb4_0900_ai_ci				☐
5	hire_date	DATE		☐	☐	☐	No default						☐
6	salary	DECIMAL	10,2	☐	☐	☐	No default						☐
7	created_at	TIMESTAMP		☐	☒	☐	CURRENT_TIMEST...						☐
8	updated_at	TIMESTAMP		☐	☒	☐	CURRENT_TIMEST...						☐